Can the Rydberg Formula Be Used to Accurately Predict Experimental Balmer Hydrogen Wavelengths?

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In this project for Introduction to Programming, I compared hydrogen transition wavelengths that were obtained experimentally with wavelengths calculated using the Rydberg formula. The research question this project aimed to answer was:

Can the Rydberg formula be used to accurately predict experimental Balmer hydrogen wavelengths?

Physical Context

When hydrogen transitions from a higher energy state to a lower energy state, it releases a photon whose energy can be measured in order to determine the transition wavelength of hydrogen. The energy difference between the two states determines the wavelength:

$$\Delta E = \frac{hc}{\lambda}. \tag{1}$$

On the other hand, there is a way to predict the transition wavelength mathematically. Using the Schrödinger equation to determine the hydrogen energy levels gives a dependence of

$$E_n \propto -\frac{1}{n^2}, \tag{2}$$

which is the same dependence seen in the Rydberg formula:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right). \tag{3}$$

Using this formula, we can calculate a prediction of the transition wavelength, where m is the lower energy state and n is the higher energy state from which the hydrogen transitions.

Dataset

The experimental data used for this project was taken from the NIST Atomic Spectra Database and filtered for hydrogen lines between 300–760 nm. Only wavelengths measured in vacuum were used for this project.

Another important filter was the Balmer series, as the Rydberg formula was used to predict transition wavelengths between principal quantum numbers. Therefore, data containing fine-structure components such as $2s$ or $2p$ were excluded.

Results

The results showed a very accurate prediction of the hydrogen transition wavelengths. With a mean percentage error of only 0.00089%, the predicted values were very close to the experimental values. The differences in wavelength can be observed in the table below.

n_upper	obs_wl_vac(nm)	predicted_wavelength_nm	difference_nm	percentage_error
9	383.6485	383.651081	−0.002581	0.000673
8	389.0166	389.019038	−0.002438	0.000627
7	397.1198	397.123601	−0.003801	0.000957
6	410.2892	410.293517	−0.004317	0.001052
5	434.1692	434.173034	−0.003834	0.000883
4	486.2710	486.273798	−0.002798	0.000575
3	656.4600	656.469627	−0.009627	0.001466

Figure 1: Comparison of expected and experimental wavelength of hydrogen including percentage error

The normalized difference was calculated using the experimental uncertainty:

$$\text{Normalized difference} = \frac{\lambda_{\text{observed}} - \lambda_{\text{predicted}}}{\sigma_{\text{observed}}}. \tag{4}$$

This showed that, while the predictions were very accurate, several differences were large compared with the experimental uncertainties. This is mostly due to the experimental uncertainties listed by NIST being very small.

	n_upper	difference_nm	unc_obs_wl	normalized_difference
0	9	−0.002581	0.0006	−4.300899
4	8	−0.002438	0.0006	−4.063408
5	7	−0.003801	0.0006	−6.335562
8	6	−0.004317	0.0006	−7.194480
9	5	−0.003834	0.0006	−6.389220
14	4	−0.002798	0.0050	−0.559511
18	3	−0.009627	0.0030	−3.208900

Figure 2: Normalized differences between observed and predicted hydrogen wavelength

The plotted results show the same pattern, as there is an almost perfect linear relationship between the observed and calculated values, with very little relative difference between the wavelengths.

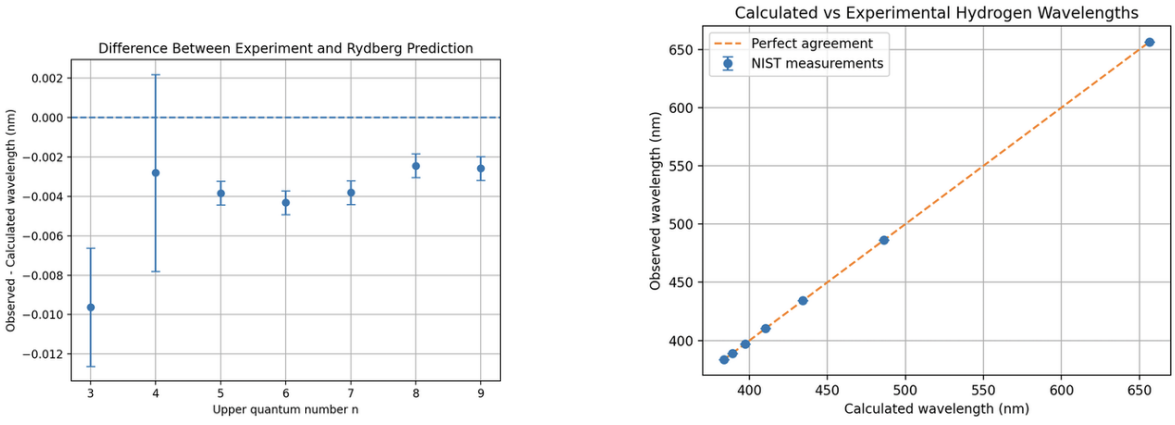


Figure 3: Residual wavelength difference vs upper quantum number n (left), Calculated vs Experimental Hydrogen Wavelength (right)

Sources

- Kramida, A., Ralchenko, Yu., Reader, J., and NIST ASD Team (2024). *NIST Atomic Spectra Database*, version 5.12. National Institute of Standards and Technology, Gaithersburg, MD. DOI: 10.18434/T4W30F. Accessed 7 October 2026.
- Vos, K. (2026). *Quantum Mechanics, Lecture 4.* Maastricht Science Programme, Maastricht University. Course lecture.